

Customer Shopping Behavior Analysis

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1. Project Overview This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in the Review Rating column.

3. Exploratory Data Analysis & Modeling (Python) The dataset was prepared, cleaned, and modeled using Python before database integration:

- **Data Loading & Inspection:** Loaded via Pandas, utilizing `info()` and `describe()` to verify dimensions and check data distributions.

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   object
3   Item Purchased        3900 non-null   object
4   Category              3900 non-null   object
5   Purchase Amount (USD) 3900 non-null   int64
6   Location              3900 non-null   object
7   Size                  3900 non-null   object
8   Color                 3900 non-null   object
9   Season                3900 non-null   object
10  Review Rating         3863 non-null   float64
11  Subscription Status    3900 non-null   object
12  Shipping Type          3900 non-null   object
13  Discount Applied       3900 non-null   object
14  Promo Code Used        3900 non-null   object
15  Previous Purchases     3900 non-null   int64
16  Payment Method         3900 non-null   object
17  Frequency of Purchases 3900 non-null   object
dtypes: float64(1), int64(4), object(13)
```

4. Missing Data Handling: Imputed missing values in the `review_rating` column using the median rating of each respective product category.

```
df.isnull().sum()
[8] ✓ 0.0s Python

... Customer ID      0
Age                0
Gender            0
Item Purchased    0
Category          0
Purchase Amount (USD) 0
Location          0
Size              0
Color             0
Season            0
Review Rating     37
Subscription Status 0
Shipping Type     0
Discount Applied  0
Promo Code Used   0
Previous Purchases 0
Payment Method    0
Frequency of Purchases 0
dtype: int64
```

Column Standardization: Cleaned headers into standard snake_case formatting.

Feature Engineering: Created the age_group binning column and transformed purchase frequencies into numeric days.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
[25] ✓ 0.0s Python

Success .columns
[12] ✓ 0.0s Python

... Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases'],
        dtype='object')

#creating a new column age_group

labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
[13] ✓ 0.0s Python
```

Database Integration: Connected the workflow directly to a relational database to host the clean table for structured querying.

```
import pandas as pd
from sqlalchemy import create_engine

# 1. Create your MySQL database connection engine
# Replace username, password, host, port, and database_name with your actual MySQL credentials
engine = create_engine('mysql+pymysql://root:password@localhost:3306/customer_db')

# 2. Transfer the cleaned DataFrame directly to the MySQL table
df.to_sql(
    name='cleaned_customer',
    con=engine,
    if_exists='replace', # Options: 'fail', 'replace', or 'append'
    index=False         # Keeps Pandas from saving row numbers as a column
)
[24] ✓ 0.6s Python

... 3900
```

4. Data Analysis using SQL (Business Transactions)

Structured queries were executed against the database to solve key business objectives, split between core operational metrics and advanced custom analytics:

Custom Analytics Insights.

- **Top Revenue-Generating Locations:** Evaluated geographic performance to isolate top markets and trace average customer ages.

location	total_revenue	avg_customer_age
Montana	5784	43.8
Illinois	5617	43.3
California	5605	42.7
Idaho	5587	43.3
Nevada	5514	43.6

Payment Method Preferences: Tracked transaction volumes and total revenues across gateways like PayPal and credit cards.

payment_method	total_transactions	total_revenue
Credit Card	671	40310
PayPal	677	40109
Cash	670	40002
Debit Card	636	38742
Venmo	634	37374
Bank Transfer	612	36544

Seasonal Category Performance: Cross-referenced seasons and categories to optimize seasonal inventory flows.

season	category	total_revenue
Fall	Clothing	26220
Fall	Accessories	19874
Fall	Footwear	8665
Fall	Outerwear	5259
Spring	Clothing	27692
Spring	Accessories	17007
Spring	Footwear	9555
Spring	Outerwear	4425
Summer	Clothing	23078
Summer	Accessories	19028
Summer	Footwear	9393
Summer	Outerwear	4278
Winter	Clothing	27274
Winter	Accessories	18291
Winter	Footwear	8480
Winter	Outerwear	4562

Discount Impact on Review Ratings: Measured whether discounted items experience variances in customer satisfaction scores.

discount_applied	avg_review_rating	total_purchases
Yes	3.74	1677
No	3.76	2223

Revenue by Gender: Compared total spend metrics between male and female cohorts.

group by gender

gender	revenue
Male	157890
Female	75191

Subscribers vs. Non-Subscribers: Analyzed average basket sizes and overall revenue generation for active subscribers.

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

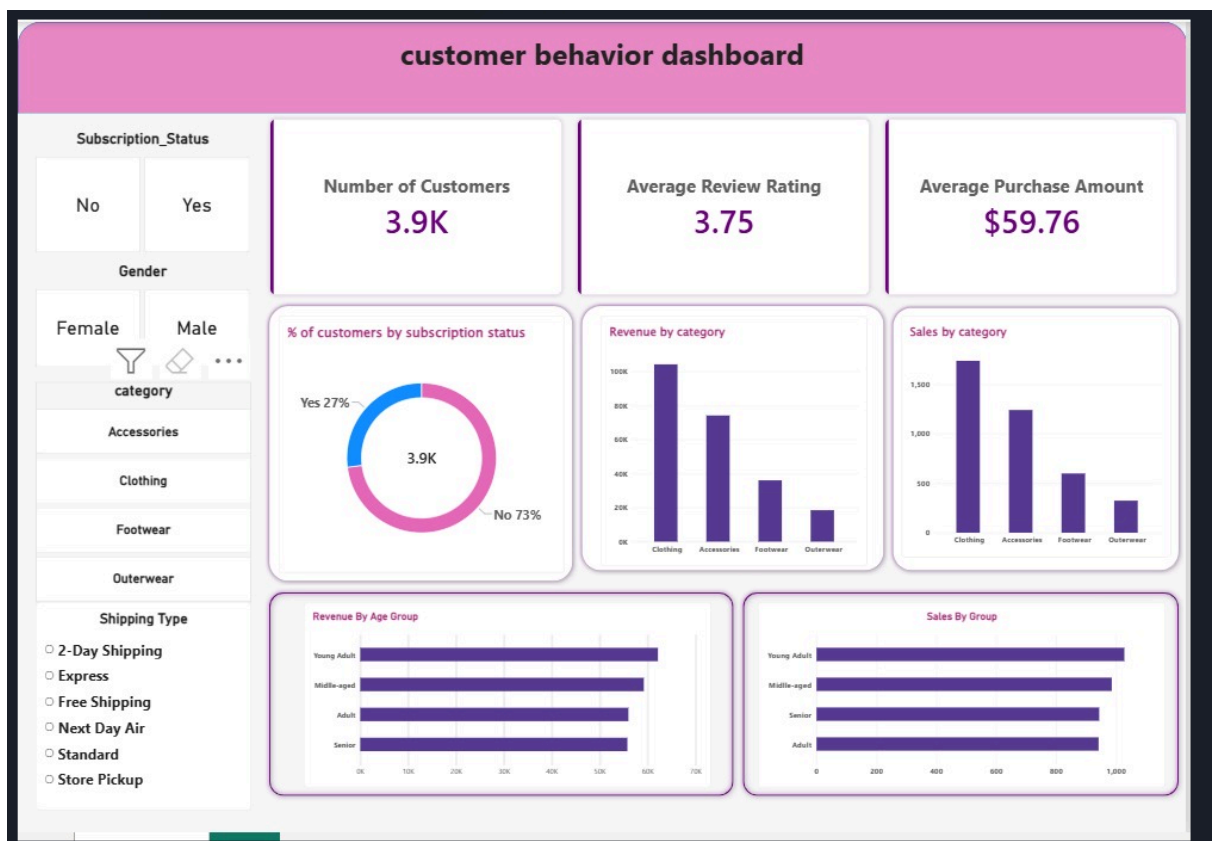
Customer Segmentation: Grouped buyers into New, Returning, and Loyal tiers based on historical purchase counts.

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

Top Products per Category: Extracted top-ranked items using window functions (ROW_NUMBER()) partitioned by category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

5. Visualization & Interactive Dashboard (Power BI) An interactive Power BI dashboard was built to translate the underlying database metrics into executive-ready visual components.



Core KPIs: High-level metric cards track Total Customers (3.9K), Average Purchase Amount (\$59.76), and Average Review Rating (3.75).

Behavioral Breakdowns: Donut charts illustrate subscription distribution (27% Yes vs. 73% No), while column/bar charts outline category sales, revenue performance by age brackets, and geographic concentrations.

6. Business Recommendations

- Optimize Geographic & Regional Campaigns: Focus marketing capital on top-performing locations identified in location analytics.
- Boost Subscription Adoption: Introduce tiered rewards for non-subscribers to shift them toward high-retention subscription models.
- Refine Discount Strategies: Evaluate the margin trade-offs of heavily discounted product lines to ensure long-term profitability.
- Leverage Lifecycle Segmentation: Deploy targeted retention workflows customized specifically for Returning and Loyal customer tiers.